

How the Congress-Trade Agent Works

This document explains, in plain language, what the automated trading agent does, how it picks stocks, how it enters and exits positions, and what safety limits are in place. All trading happens on an Alpaca paper-trading account — no real money is at risk.

1. What the agent does

Every 15 minutes during US market hours, the agent looks up recent stock trades disclosed by members of the US Congress (via the Quiver Quantitative API). If a politician bought a stock that matches our criteria, the agent mirrors the buy in the paper account with a fixed \$500 position. It then attaches a 5% trailing-stop sell order that automatically closes the position if the price drops 5% from its peak.

2. How signals are picked

A disclosure becomes an actionable signal only if it passes all of these filters:

- Transaction type is Purchase (sales, exchanges, and partial sales are ignored).
- Asset is a regular US-listed stock (options, bonds, mutual funds, and ETFs are skipped).
- The disclosed dollar range is at least \$15,001 (the smallest SEC disclosure bucket is skipped).
- The disclosure was filed within the last 7 days.
- The ticker is tradable on Alpaca.

3. How trades are executed

- Position size: \$500 per new signal, placed as a notional market order (fractional shares).
- Max open positions: 10 at a time.
- Max total exposure: \$10,000 deployed at any moment.
- If you hold a stock already — agent or manual — the agent will not double-buy it.

4. How positions exit

Once the buy fills, the agent attaches a 5% trailing-stop sell order. A trailing stop follows the stock's peak price: if the price keeps rising, the stop rises with it; if the price drops 5% from the highest point reached, the stop triggers and sells. The

trailing stop is held server-side by Alpaca, so it works even when the agent is not running.

5. How duplicates are avoided

Each disclosure has a deterministic 20-character ID. When the agent places a buy, it tags the Alpaca order with "buy-". Before acting on a new signal, it checks the last 60 days of Alpaca orders — if it already placed a buy for this disclosure, it skips. This works across agent runs without needing a local database.

6. Which positions get trailing stops

The trailing stop is only attached to positions the agent itself bought (matched by the "buy-" prefix on the client order ID). Positions you bought manually in the Alpaca dashboard are left untouched.

7. What you will see

- Email on every buy placed.
- Email when a trailing stop is attached.
- Email when a trailing stop triggers and the position sells.
- Portfolio summary on every GitHub Actions run (account equity, positions, pending orders, recent fills).
- Daily digest email at 4:30 PM ET on trading days.

8. Safety rails

- Only runs on an Alpaca paper account — not real money.
- Skips all orders when the US market is closed (no weekend trading).
- Hard caps on position count, position size, and total exposure.
- All credentials are stored as GitHub Actions encrypted secrets, never in the code.

Disclaimer: This is an educational project. Published research on "follow Congress" strategies is mixed — some politicians show alpha, many do not. Results on a paper account do not translate directly to live trading.